Ouster OS0 Case Study: Errors and Resolution

BIG IDEA

Small marks, steady answers and a true ruler are three different things.

resolution: how small the marks are

accuracy: is the ruler the right length? (stretched!)

precision: how much your answers wiggle

THE IDEA

A datasheet is like the label on a box of crayons: it tells how many colors, how sharp the tips are and how long they last. To know whether you can draw a tiny ant or only a big elephant, you do a little math with those numbers. That is what this module does for one real laser scanner.

REMEMBER

- ✓ **Resolution** is how small the marks are.
- ✓ **Precision** is how much your answer wiggles if you measure the same thing ten times.
- ✓ **Accuracy** is whether the ruler itself is the right length. Tiny marks can still be on a stretched ruler!

TRY THIS

Measure the width of a book five times with a ruler. Write each answer down. How much do they wiggle?